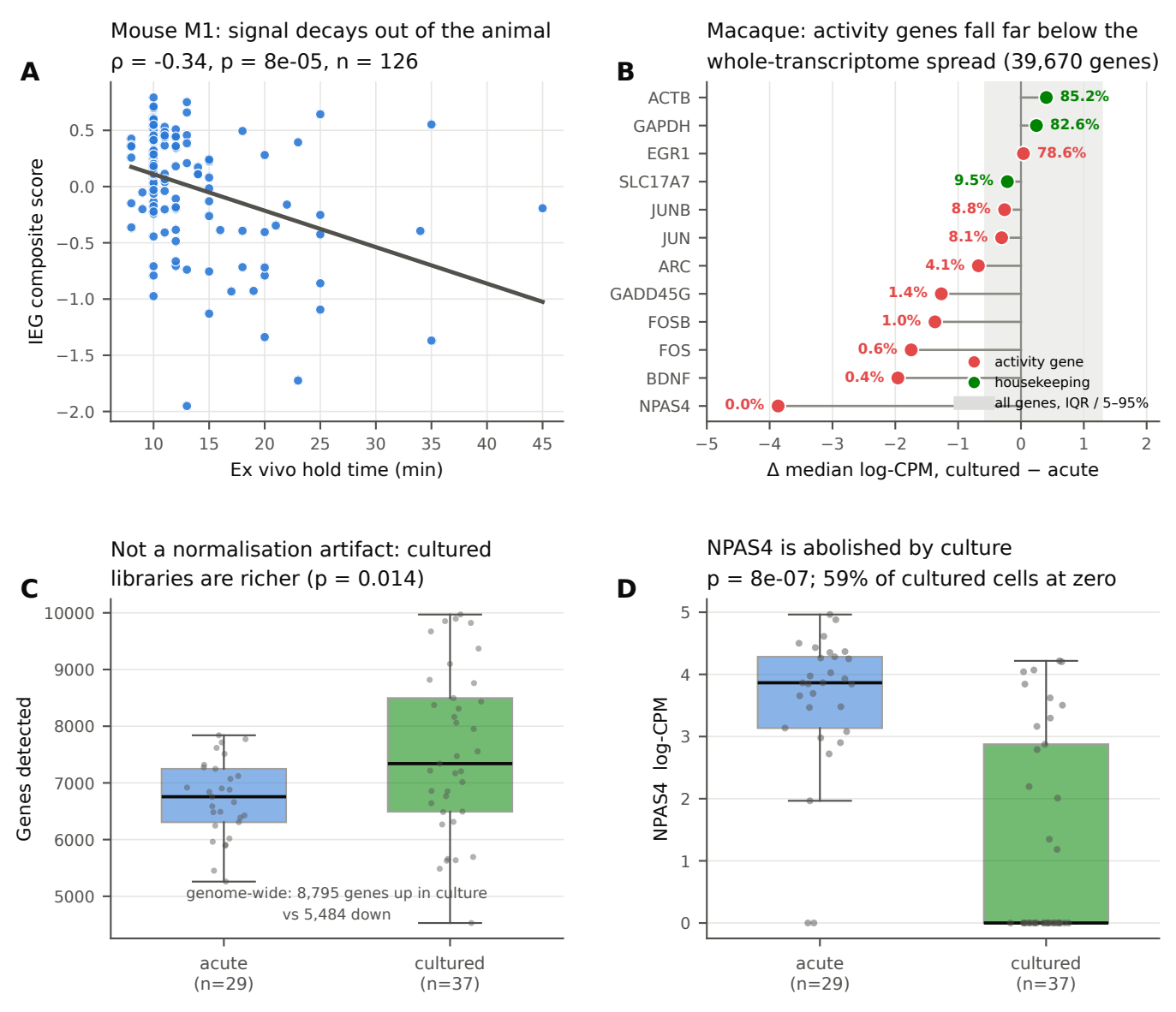
A
Mouse M1: signal decays out of the animal
ρ = -0.34, p = 8e-05, n = 126
IEG composite score
Ex vivo hold time (min)
B
Macaque: activity genes fall far below the
whole-transcriptome spread (39,670 genes)
ACTB
GAPDH
EGR1
SLC17A7
JUNB
JUN
ARC
GADD45G
FOSB
FOS
BDNF
NPAS4
85.2%
82.6%
78.6%
9.5%
8.8%
8.1%
4.1%
1.4%
1.0%
0.6%
0.4%
0.0%
activity gene
housekeeping
all genes, IQR / 5–95%
Δ median log-CPM, cultured − acute
C
Not a normalisation artifact: cultured
libraries are richer (p = 0.014)
Genes detected
genome-wide: 8,795 genes up in culture
vs 5,484 down
acute
(n=29)
cultured
(n=37)
D
NPAS4 is abolished by culture
p = 8e-07; 59% of cultured cells at zero
NPAS4 log-CPM
acute
(n=29)
cultured
(n=37)